

Let $E: y^2 = x^3 + x + 6$, where the arithmetic is defined over $\mathbb{Z}_{11}$ ($\mathbb{Z}_{11}$ is also written as $GF(11)$). The group that this defines is denoted as $E(\mathbb{Z}_{11})$ and has 13 elements, which is denoted as $\#E(\mathbb{Z}_{11}) = 13$.

Set generator $\alpha = (2,7)$, then we compute the following sequence of points.

$$\begin{array}{lll} \alpha = (2,7) & 2\alpha = (5,2) & 3\alpha = (8,3) \\ 4\alpha = (10,2) & 5\alpha = (3,6) & 6\alpha = (7,9) \\ 7\alpha = (7,2) & 8\alpha = (3,5) & 9\alpha = (10,9) \\ 10\alpha = (8,8) & 11\alpha = (5,9) & 12\alpha = (2,4) \end{array}$$

Example 1

Given that $3 \cdot \alpha = (8,3)$, compute $2 \cdot (3 \cdot \alpha) = 3 \cdot \alpha + 3 \cdot \alpha$

$$\begin{aligned} \lambda &= \frac{3 \cdot (8)^2 + 1}{2 \cdot (3)} = \frac{3 \cdot 64 + 1}{6} = \frac{6}{6} = 1 \pmod{11} \\ x &= (1)^2 - 2 \cdot (8) = 1 - 16 = -15 = 7 \pmod{11} \\ y &= (8 - 7) \cdot 1 - 3 = -2 = 9 \pmod{11} \\ \therefore 2 \cdot (3 \cdot \alpha) &= (7,9) \end{aligned}$$

Observe that $2 \cdot (3 \cdot \alpha) = 6 \cdot \alpha = (7,9)$

It can be shown that we also have $3 \cdot (2 \cdot \alpha) = 6 \cdot \alpha = (7,9)$.

Example 2

Given that $12 \cdot \alpha = (2,4)$, compute $12 \cdot \alpha + \alpha = (2,4) + (2,7)$

$$\lambda = \frac{7 - 4}{2 - 2} = \frac{3}{0}$$

This calculation shows that $12 \cdot \alpha + \alpha = \mathcal{O} = 13 \cdot \alpha$, which is the order of the point α (where $\mathcal{O}$ is the point at infinity on the elliptic curve).

Furthermore, it can be shown that $14 \cdot \alpha = \alpha$, because $\alpha + 13 \cdot \alpha = \alpha + \mathcal{O} = \alpha$.

In other words, the sequence of points repeats after we reach the point at infinity.

Example 3

Given that $7 \cdot \alpha = (7,2)$, compute $2 \cdot (7 \cdot \alpha) = (7,2) + (7,2)$

$$\begin{aligned} \lambda &= \frac{3 \cdot (7)^2 + 1}{2 \cdot (2)} = \frac{148}{4} = \frac{5}{4} = 5 \cdot 3 = 15 = 4 \pmod{11} \\ x &= 4^2 - 2 \cdot (7) = 16 - 14 = 2 \pmod{11} \\ y &= (7 - 2) \cdot 4 - 2 = 20 - 2 = 18 = 7 \pmod{11} \\ \therefore 2 \cdot (7 \cdot \alpha) &= (2,7) \end{aligned}$$

Observe that $2 \cdot (7 \cdot \alpha) = 14 \cdot \alpha = \alpha + 13 \cdot \alpha = \alpha = (2,7)$ because α has order 13.

(Note that $\frac{5}{4} = 5 \cdot (4^{-1}) = 5 \cdot 3$ because $4^{-1} = 3$, which follows from $3 \cdot 4 = 1 \pmod{11}$)

Example 4

Compute $5 \cdot (2,7)$

Begin by computing $2 \cdot (2,7) = (2,7) + (2,7)$

$$\lambda = \frac{3 \cdot (2)^2}{2 \cdot (7)} = \frac{12 + 1}{14} = \frac{13}{14} = \frac{2}{3} = 2 \cdot 4 = 8 \pmod{11}$$

$$x = (8)^2 - 2 \cdot (2) = 5 \pmod{11}$$

$$y = (2 - 5) \cdot 8 - 7 = 2 \pmod{11}$$

$$\therefore 2 \cdot (2,7) = (5,2)$$

(Note that $\frac{2}{3} = 2 \cdot (3^{-1}) = 5 \cdot 4$ because $3^{-1} = 4$, which follows from $3 \cdot 4 = 1 \pmod{11}$)

Next compute $2 \cdot (5,2) = (5,2) + (5,2)$

$$\lambda = \frac{3 \cdot (5)^2 + 1}{2 \cdot (2)} = \frac{76}{4} = \frac{10}{4} = 10 \cdot 3 = 8 \pmod{11}$$

$$x = (8)^2 - 2 \cdot (5) = 64 - 10 = 54 = 10 \pmod{11}$$

$$y = (5 - 10) \cdot 8 - 2 = -40 - 2 = -42 = 2 \pmod{11}$$

$$\therefore 2 \cdot (5,2) = (10,2)$$

Also, we have $2 \cdot (2 \cdot (2,7)) = 4 \cdot (2,7) = (10,2)$.

(Note that $\frac{10}{4} = 10 \cdot (4^{-1}) = 10 \cdot 3$ because $4^{-1} = 3$, which follows from $3 \cdot 4 = 1 \pmod{11}$)

Finally compute $(10,2) + (2,7)$

$$\lambda = \frac{7 - 2}{2 - 10} = \frac{5}{-8} = \frac{5}{3} = 5 \cdot 4 = 20 = 9 \pmod{11}$$

$$x = (9)^2 - 10 - 2 = 81 - 12 = 69 = 3 \pmod{11}$$

$$y = (10 - 3) \cdot 9 - 2 = 63 - 2 = 61 = 6 \pmod{11}$$

$$\therefore (10,2) + (2,7) = (3,6) = 5 \cdot (2,7)$$

(Note that $\frac{5}{4} = 5 \cdot (4^{-1}) = 5 \cdot 3$ because $4^{-1} = 3$, which follows from $3 \cdot 4 = 1 \pmod{11}$)

So, to summarise what we did, we computed $5 \cdot (2,7)$ by adding and doubling points as

$$2 \cdot (2 \cdot (2,7)) + 1 \cdot (2,7)$$

$$4 \cdot (2,7) + 1 \cdot (2,7)$$

$$5 \cdot (2,7) = (3,6)$$

Essentially, this is because $2 \cdot 2 + 1 = 5$.